

Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Alessandro Gori*

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Abstract

[To be continued. . .]

- Appendix on s -wave SC ghost simulations on HM
- Set ComputerModern font in all CairoMakie figures

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*a.gori23@studenti.unipi.it / nepero27178 @ GitHub

List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ -wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
PH	Particle-hole
PHS	Particle-hole symmetry
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

Chapter 1

Introduction to the Extended Hubbard Model

A major breakthrough in Condensed Matter Physics was the discovery of superconductivity in two-dimensional copper oxide based compounds (often referred to as “cuprates”). Until 1986, when Bednorz and Muller found evidence of a superconducting transition in Ba-La-Cu-O compounds at a critical temperature (T_c) around 30 K [5], superconductivity had been observed up to few Kelvins. The new class of copper oxides rapidly took over in the field of high- T_c superconductivity, increasing critical temperatures to around 130 K at ambient pressure. Today, the highest observed value is $T_c \approx 151$ K at ambient pressure on the cuprate $\text{HgBa}_2\text{Ca}_2\text{Cu}_3\text{O}_{8+\delta}$ [6]. It is worth mentioning that this result was obtained in recent months after almost 30 years from the previous record.

[Figure: superconducting transition temperatures]

Cuprates are very rich materials, with a complicated and yet not fully understood phase diagram. Undoped copper oxides show an antiferromagnetic ground state, which is disrupted both by electronic and hole doping. At low temperatures, anisotropic superconductivity is established

[AF ordering at low doping; Mott transition; pseudogap and strange metal; unconventional superconductivity]

1.1 Copper oxides

The crystal structure of copper oxide superconductors is given by a stacking of 2D CuO_2 sheets separated by inert layers of insulating materials. These intermediate layers act as reservoirs of electrons (or holes) and dope the active CuO_2 sheets [29]. Due to the presence of these intermediate inert layers, the CuO_2 sheets are weakly coupled to each other, and can be studied independently. In these sheets, copper and oxygens are arranged as a checkerboard lattice: oxygen atoms form a square lattice, and in the center of each cell is placed a copper atom. At null doping, due to the high electronegativity of oxygens, copper atoms are ionised and left with one hole in their d shell, while oxygen atoms reach a closed shell configuration. Therefore, at low energy the kinetics of electrons is given by the hopping among the d orbitals of Cu^{2+} ions, arranged in a planar square lattice [Lee 2006].

Electrons repel each other through Coulomb interaction. The simplest model capable of describing this kinetics and electron-electron repulsion is the single-band Hubbard model [Hubbard 1963],

$$\hat{H}_{\text{HM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \quad \text{where } t, U > 0, \quad (1.1)$$

being t the hopping energy and U the Coulomb repulsion. Sum is intended on the spin index $\sigma \in \{\uparrow, \downarrow\}$ and the sites i, j of a planar square lattice $\mathcal{L}$. The notation $\langle ij \rangle$ indicates nearest neighbours (NNs). Note that the Coulomb repulsion is limited to be a on-site interaction. It is known that the Hubbard model captures the low-energy properties of copper oxides [Zhang 1988, Lee 2006].

1.2 Mott physics

Conventional band theory describes the electronic properties of materials on the assumption that electron-electron interaction is negligible. Within this framework, a material forms a band insulator whenever the conduction band fills up. Due to Pauli principle, electrons fill all states of the band and have no room left for low-energy scattering, hence the filled band is completely ineffective for conduction. On the other hand, if the conduction band is partially filled, the material is a metal and conducts electricity [29]. In general, band theory predicts a metallic behaviour if the number of electrons per unit cell is odd; an insulating behaviour if it is even.

In undoped copper oxides, the conduction band is not completely filled. Indeed, each Cu^{2+} ion accommodates one hole in its most external orbital. Band theory would predict a metallic nature for these compounds, but the observed state is insulating. The reason for this discrepancy lies in the fact that electron-electron interaction is not negligible. Cuprates are Mott insulators, not band insulators: their insulating properties do not derive from a single-particle band being filled up and separated by a gap from another band at higher energy; instead, from an amplification of Coulomb repulsion that “freezes” electrons on the lattice sites [7]. One can picture this situation as a sort of traffic jam, where electrons do not move because the energetic cost of having two electrons on the same site would be too high [Zaneen 2015]. This produces an insulating behaviour: electrons are completely localised and conductivity is low.

[Figure: traffic jam and Mott insulation]

The Hubbard model describes naturally Mott physics. Consider for instance the two-sites Hubbard model,

$$\hat{H} = -t \left(\hat{c}_{1\uparrow}^\dagger \hat{c}_{2\uparrow} + \hat{c}_{2\downarrow}^\dagger \hat{c}_{1\downarrow} \right) + U \left(\hat{n}_{1\uparrow} \hat{n}_{1\downarrow} + \hat{n}_{2\uparrow} \hat{n}_{2\downarrow} \right),$$

where the subscript indicates the site index. In the strong coupling limit $U \gg t$, we can treat the kinetic terms perturbatively. Indeed, let E_i be the energy given by having i electrons on one site. At order zero in perturbation theory, the difference in energy between the state with one site doubly occupied and the other empty, and the state with two singly occupied sites is

$$E_2 + E_0 - 2E_1 = U.$$

The zeroth order ground-state, then, is a combination of states where both sites are singly occupied. Insertion of the kinetic terms as a perturbative correction does not change the scenario (see App. A for details).

In the Hubbard model, the transition from a metallic state to a Mott insulator is controlled by the strength of the Coulomb repulsion U , and impacts visibly the density of states (DoS), which undergoes significant changes [Kotliar 2004]. Let us sketch the general behaviour: to start, let $U = 0$ and consider a lattice with one electron per site (half-filling). Electrons are free and do not interact, therefore move freely across the lattice. Due to Pauli exclusion principle, they form a Fermi sea that occupies the lower half a metallic band with a width W . As we increase the value of U , it becomes increasingly harder for electrons to move. When U becomes sufficiently large (with respect to W), the electronic density of states splits in two bands. The lower band, filled, accommodates all completely localised electrons; in the upper band, empty, are all the states where a site gets doubly occupied. This splitting is of order $\sim U$. At intermediate values of U , the DoS presents a peculiar three-peaks shape, which is the result of a mixture of the two Mott bands, and a residual of the original free band around the Fermi energy. This is shown in Fig. [Missing].

[Figure: Hubbard bands]

One of the signatures of a Mott transition, thus, is the flattening of the bare single-particle bands.

1.3 Antiferromagnetism

It is widely known that the ground-state of undoped copper oxides exhibits antiferromagnetic ordering [7]. The insurgence of this phase can be connected to Mott physics in the Hubbard model. Indeed, in the strong coupling regime $U \gg t$, the Hubbard hamiltonian is effectively approximated by the Heisenberg hamiltonian with antiferromagnetic coupling [Marston 1989],

$$\hat{H}_{\text{HM}} \simeq \frac{4t^2}{U} \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j,$$

where $\hat{\mathbf{S}}_i$ is the spin operator on site i .

Antiferromagnetism is established by taking advantage of the “superexchange mechanism”.

1.4 d -wave superconductivity

It is known that injection of charge (electrons or holes) in copper oxide compounds disrupts antiferromagnetic ordering and establishes a variety of peculiar phases [7]. Among these, the high temperature superconducting phase is significantly different from the conventional superconducting phase described by Bardeen-Cooper-Schrieffer (BCS) theory.

In BCS theory, couples of electrons are glued together by an effective attraction that is mediated by lattice phonons. This mechanism is referred to as “Cooper pairing”. The resulting state is a condensate of Cooper pairs, with an energy spectrum that is fully gapped and thus no low-energy excitations. The gap $\Delta_{\mathbf{k}}$, as a function of momentum, is isotropic (s -wave) and represents the energy cost of breaking a Cooper pair glued together by phononic interactions.

In copper oxides, superconductivity is established with significantly different features. Most notably, the superconducting gap is not isotropic but changes sign under rotation of $\pi/2$ on the plane. Because of this, we talk about d -wave superconductivity. A consequence is the appearance of gapless quasiparticle excitations in the energy spectrum. Indeed, a d -wave gap has nodal lines (continuous sets of points, in reciprocal space, where it is null) which, in a square lattice geometry, result in a set of nodes around which dispersion is linearised (Dirac cones) [28].

In the phase diagram of copper oxides, the d -wave superconducting region “emerges” from the antiferromagnetic region, as doping is increased at low temperature. The relation between Mott physics and d -wave superconductivity is yet not fully understood. It has been proposed that the same Mott-induced spin fluctuations that lead to the formation of the antiferromagnetic state could act as an effective Cooper pairing [28] [Monod 1986, Miyake 1986, Scalapino 1986]. This problem is still open and is subject of a great effort.

Appendix A

Superexchange and virtual hopping in Hubbard lattices

A key mechanism in AF phase formation in Hubbard lattice is superexchange. The AF phase is stabilized by spin fluctuations and second-order virtual hopping. The mechanism becomes clear enough by considering a 2-sites Hubbard toy model.

A.1 Virtual hopping in the 2-sites Hubbard lattice

Consider the toy model:

$$\hat{H} = -t \left\{ \hat{c}_{1\uparrow}^\dagger \hat{c}_{2\uparrow} + \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\downarrow} + \text{h.c.} \right\} + U \{ \hat{n}_{1\uparrow} \hat{n}_{1\downarrow} + \hat{n}_{2\uparrow} \hat{n}_{2\downarrow} \}$$

with $i = 1, 2$ the site index. The two sites are represented in Fig. A.1. The two competing processes are:

1. electrons inter-sites hopping with amplitude $-t$;
2. local repulsion $+U$, acting when two anti-aligned electrons reside on the same site;

For an half-filled system, the Hilbert space is six-dimensional. I use the notation $|n_{1\uparrow}n_{1\downarrow}n_{2\uparrow}n_{2\downarrow}\rangle$ to indicate the six computational basis states:

$$\begin{array}{lll} |\psi_1\rangle \equiv |1010\rangle & |\psi_3\rangle \equiv |1001\rangle & |\psi_5\rangle \equiv |0011\rangle \\ |\psi_2\rangle \equiv |1100\rangle & |\psi_4\rangle \equiv |0110\rangle & |\psi_6\rangle \equiv |0101\rangle \end{array}$$

For example, the top panel of Fig. A.1 shows state $|\psi_2\rangle$.

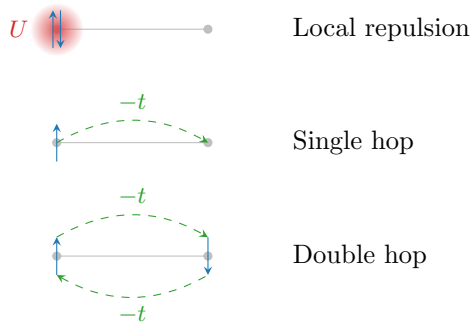


Figure A.1 | Two sites Hubbard model.

Structure	Eigenstate	Energy
Spin-1/2 singlet	$\frac{ \phi_3\rangle - \phi_4\rangle}{\sqrt{2}}$	$E^- = \frac{U}{2} - \sqrt{\frac{U^2}{4} + 4t^2}$
Spin-1/2 triplet	$ \phi_1\rangle, \frac{ \phi_3\rangle + \phi_4\rangle}{\sqrt{2}}, \phi_6\rangle$	0
	$\frac{ \phi_2\rangle - \phi_5\rangle}{\sqrt{2}}$	U
	$\frac{ \phi_2\rangle + \phi_5\rangle}{\sqrt{2}}$	$E^+ = \frac{U}{2} + \sqrt{\frac{U^2}{4} + 4t^2}$

Table A.1 | List of exact eigenstates and relative energies for the 2-sites half-filled Hubbard model.

A.1.1 Exact solution of the half-filled model

The hamiltonian matrix is directly evaluated in this basis

$$H_{ij} = \langle \psi_i | \hat{H} | \psi_j \rangle \implies H = \begin{bmatrix} 0 & & & & & \\ & U & -t & -t & & \\ & -t & & & -t & \\ & -t & & & -t & \\ & & -t & -t & U & \\ & & & & & 0 \end{bmatrix}$$

Empty slots in the matrix stand for zeros. Evidently the states $|\psi_1\rangle$ (both up) and $|\psi_6\rangle$ (both down) are zero-energy eigenstates. These states cannot realize electrons hopping because of Pauli principle. The internal 4×4 matrix is readily diagonalized by the means of a change of basis V ,

$$V \begin{bmatrix} U & -t & -t & \\ -t & & & -t \\ -t & & & -t \\ & -t & -t & U \end{bmatrix} V^\dagger = \begin{bmatrix} E^- & & & \\ & 0 & & \\ & & U & \\ & & & E^+ \end{bmatrix}$$

Tab. A.1 shows the eigenvectors and relative eigenvalues obtained from diagonalization. The ground-state is the singlet state,

$$\frac{|\phi_3\rangle - |\phi_4\rangle}{\sqrt{2}} = \frac{|1010\rangle - |0101\rangle}{\sqrt{2}} = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}$$

of energy

$$E^- = \frac{U}{2} - \sqrt{\frac{U^2}{4} + 4t^2} \simeq -\frac{4t^2}{U}$$

the latter equality being true if $U \gg t$ (strong repulsion limit). The singlet state pairs with a spatially-symmetric (nodeless) wavefunction. The entire triplet (second row of Tab. A.1) remains at zero energy. Excited states are anti-symmetrized and symmetrized version of the polarized states $|\phi_1\rangle$ and $|\phi_6\rangle$.

A.1.2 Virtual hopping

The key feature of the singlet state is the one represented in the bottom panel of Fig. A.1: if the two electrons occupy separate sites and are anti-aligned, both “see” the other site as empty, thus free to hop to. [To be continued...]

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